Constructing the Real Numbers from Cauchy Sequences

Chapter 4 ended at a hole. The Newton sequence $2, 3/2, 17/12, 577/408, \ldots$ of Chapter 3 is Cauchy in $\mathbb{Q}$, and no rational is its limit (Theorem 4.6.1). Its terms crowd around a place where $\mathbb{Q}$ has no element. This chapter does not search for a number in the hole. It **takes the sequence itself as the number**. Many sequences crowd around the same hole: the decimal truncations $1, 7/5, 141/100, \ldots$ approach it from below and never meet the Newton sequence. So two Cauchy sequences of rationals must count as one number when their difference dies out. That is an equivalence relation, and *Sets and Functions*, Section 5.2 knows what to do with one: take its classes. *A real number is a Cauchy sequence of rationals, up to a null difference.* The chapter defines the sum, product and order of these classes on representatives and proves each well defined. The laws of arithmetic are read off the laws of $\mathbb{Q}$, one index at a time. The order needs a new idea, because comparing terms does not compare classes. The chapter then places $\mathbb{Q}$ inside the new field, proves it Archimedean with $\mathbb{Q}$ dense in it, and finds there a positive number whose square is 2. Until the field exists, every tolerance, margin and threshold is rational, so nothing about real numbers is assumed on the way to building them.

Then comes completeness. A set of real numbers that is bounded above is trapped by halving intervals with rational ends; the ends form a Cauchy sequence of rationals, which is a real number, and that number is the least upper bound. So the constructed $\mathbb{R}$ is a complete ordered field. Chapter 6 shows that any two complete ordered fields are isomorphic, and *Axiomatic Set Theory* builds one from Dedekind cuts. *Analysis on the Real Line*, Chapter 1 takes the axioms of a complete ordered field as its starting point, and this chapter gives them a model. *Analytic Preliminaries*, Chapter 5 builds the complex numbers on this $\mathbb{R}$. *Analysis on the Real Line*, Chapter 5 repeats the construction for an arbitrary metric space.

What you need. Chapter 4 above all, applied with $F = \mathbb{Q}$ and, once Section 5.3 has made $\mathbb{R}$ an ordered field, with $F = \mathbb{R}$; Chapters 2 and 3; Chapter 1's arithmetic and order of $\mathbb{N}$; *Sets and Functions*, Chapters 2–6, chiefly Section 5.2; and *Logic and Proof*, Chapters 4 and 6. From Chapter 4: the rules of an ordered field and the absolute value (Proposition 4.1.2, Lemma 4.1.1, Definition 4.1.1, Proposition 4.1.3); the copy ι_F of $\mathbb{Q}$ in an ordered field (Theorem 4.2.2); the Archimedean property in its reciprocal form, the integer part and density (Proposition 4.2.3, Proposition 4.2.4, Theorem 4.2.5); null sequences, tail changes, the limit laws, null times bounded, and limits preserving $\leq$ (Definition 4.3.3, Proposition 4.3.2, Theorem 4.4.3, Proposition 4.4.1, Proposition 4.4.4); Cauchy sequences, their boundedness and algebra, the sign theorem and reciprocals (Definition 4.5.1, Proposition 4.5.2, Theorem 4.5.3, Theorem 4.5.4, Proposition 4.5.6); and the Newton sequence with no rational limit (Theorem 4.6.1). From Chapters 2 and 3: the lemma on rules defined on classes (Theorem 2.2.1); commutative rings, fields and ordered fields (Definition 2.3.1, Definition 2.4.1, Proposition 2.4.3); the order from a positive set (Proposition 2.4.1), used here a third time; least elements of sets of integers (Proposition 2.4.4); the Archimedean property of $\mathbb{Q}$ (Theorem 3.2.3); and the Newton sequence and the gaps (Example (A Rational Sequence Aimed at the Gap), Theorem 3.3.1, Theorem 3.3.2). From Chapter 1: the arithmetic of $\mathbb{N}$ and its laws (Definition 1.2.1, Theorem 1.2.3), its order and how the order meets the arithmetic (Theorem 1.3.2, Proposition 1.3.1), parity of powers (Proposition 1.3.4), and the citing rule by which induction and recursion apply to $\mathbb{N}$ (Corollary 1.4.3 (b)). From *Sets and Functions*: equivalence classes, quotient sets and the partition theorem (Sets and Functions, Definition 5.2.2, Sets and Functions, Definition 5.2.4, Sets and Functions, Theorem 5.2.1), with the method of checking a rule on representatives (Sets and Functions, Example (Method: checking that a definition on classes is well defined)); the recursion theorem

(Sets and Functions, Theorem 3.3.1); suprema, with their uniqueness and the greatest element as supremum (Sets and Functions, Definition 5.3.5, Sets and Functions, Proposition 5.3.1); and the axiom of choice (Sets and Functions, Axiom 6.1.1), named only to show that Section 5.5 does not use it. From *Logic and Proof*: induction (Logic and Proof, Axiom 4.2.1) and the negation of bounded quantifiers (Logic and Proof, Proposition 6.2.3). Some exercises also draw on the non-Archimedean field $\mathbb{Q}(t)$ of Chapter 4, the fractions of Chapter 3, countability from *Sets and Functions*, Chapter 4, and the catalogue of what a proof may not do from *Logic and Proof*, Chapter 8. Nothing from *Analysis on the Real Line* is used, and no fact about $\mathbb{R}$ is assumed.

Time. About 25 hours on the core track (all of A, one B and one C per section, the exit checks); about 35 hours for everything.

5.-1.1 Roadmap

- **Sequences That Should Converge.** The Newton sequence and the decimal truncations, defined inside $\mathbb{Q}$ by a rule, as two sequences aimed at one hole; the set $\mathcal{C}$ of Cauchy sequences of rationals; the relation $a \sim b$ when $a - b$ is null, proved an equivalence relation; the truncations equivalent to the Newton sequence; and finite changes and shifts, which never change the class.
- **The Real Numbers and Their Arithmetic.** $\mathbb{R}$ defined as the quotient set $\mathcal{C}/\!\sim$; sum, product and negative defined on representatives and proved well defined; the laws of a commutative ring read on representatives, one index at a time; and inverses through the sign theorem of Chapter 4, after the failed termwise reciprocal.
- **Positive Classes and the Order.** Why termwise comparison defines no order on classes; positive sequences, eventually above a rational margin that is named before the index; positivity as a property of the class, which makes $\mathbb{R}$ an ordered field; and termwise $\leq$ passing to classes, which strict $<$ does not.
- **The Rationals Inside the Reals.** The map $\kappa(q) = [(q)_n]$, which preserves the operations and the order and is Chapter 4's embedding $\iota_{\mathbb{R}}$; $\mathbb{R}$ Archimedean, with $\mathbb{Q}$ dense in it by Chapter 4's theorem; and the class of the Newton sequence, a positive real number whose square is 2.
- **Least Upper Bounds.** Complete ordered fields; the bound $2^k \geq k + 1$ in any ordered field; bisection with rational ends, whose right ends form a Cauchy sequence of rationals naming the supremum, so that $\mathbb{R}$ is complete; the bisection and the Newton sequence naming one number; and why no axiom of choice was needed.

5.-1.2 Named moves

The first move was taught in *Sets and Functions*; the other two are method boxes of this chapter, cited from then on.

- **Representatives, then independence** (Sets and Functions, Example (Method: checking that a definition on classes is well defined)). To define an operation or a property on classes, state it on representatives, then show that the class of the answer survives replacing each input by an equivalent one. It proves the sum, product and negative well defined in Section 5.2 and positivity a property of the class in Section 5.3. The termwise reciprocal of Section 5.2 and the termwise order of Section 5.3 fail it. Exercise B2 of Section 5.2 and Exercise C1 of Section 5.3 leave the last step, and then the last half, to you.
- **Name the margin before the index** (Example (Method: name the margin before the index)). To prove that a Cauchy sequence of rationals is positive, or that $[a] < [b]$, first name a rational margin δ from data that do not depend on n; only then ask the given sequences for thresholds at tolerances computed from δ, and take the largest. (Section

5.3; used again in the proof that $\mathbb{R}$ is Archimedean in Section 5.4 and in Steps 4 and 5 of the completeness proof in Section 5.5; Exercise B4 (d) of Section 5.3 and Exercise C3 of Section 5.4.)

- **Bisect and keep the invariant** (Example (Method: bisect and keep the invariant)). Start with a point that fails a test and one that passes it; halve by a rule that keeps one end on each side; prove the invariants and the length $D/2^k$ by induction; make the lengths small by the Archimedean property; and spend one invariant on each half of the conclusion. (Section 5.5; Exercises C1 and C2 of Section 5.5.)

5.-1.3 Threads

- **The Newton class.** Section 3.3 computes the Newton sequence and Section 4.6 proves it Cauchy with no rational limit; Section 5.1 shows the decimal truncations equivalent to it; Section 5.2 takes its class as a first real number; Section 5.4 proves that this class is positive with square 2 (Example (The Class of the Newton Sequence)); the worked example of Section 5.5 finds the same number by bisection; and Chapter 6 returns to it when it proves that every non-negative element of a complete ordered field has a square root.
- **Archimedes.** Chapter 3 proves the Archimedean property of $\mathbb{Q}$ from the construction; Section 4.2 defines it for any ordered field and derives the integer part and density; Section 5.4 proves it for $\mathbb{R}$; Step 3 of the completeness proof in Section 5.5 uses it in $\mathbb{Q}$, and density, which came from it in $\mathbb{R}$, supplies the rational ends; Chapter 6 shows that a complete ordered field is always Archimedean.
- **Choice.** *Sets and Functions*, Chapter 6 states the axiom of choice and asks every proof to name its rules; Section 5.1 defines the truncations by a rule, with no choices made; and the example on representatives in Section 5.5 shows that the completeness proof picks finitely many objects and defines the rest by recursion.

5.-1.4 Summary

The set $\mathcal{C}$ of Cauchy sequences of rationals (Definition 5.1.1) carries the relation $a \sim b$, "$a - b$ is null" (Definition 5.1.2), which is an equivalence relation (Proposition 5.1.1), and a real number is one of its classes (Definition 5.2.1). The sum, product and negative defined on representatives are well defined (Theorem 5.2.1), the laws of $\mathbb{Q}$ applied index by index make $\mathbb{R}$ a commutative ring (Theorem 5.2.2), and the sign theorem of Chapter 4 supplies inverses, so $\mathbb{R}$ is a field (Theorem 5.2.3). A Cauchy sequence of rationals is positive when its terms eventually stay above one rational margin (Definition 5.3.1); positivity belongs to the class (Proposition 5.3.1), and the positive classes make $\mathbb{R}$ an ordered field (Theorem 5.3.3), in which termwise $\leq$ passes to classes (Lemma 5.3.4). The map κ places a copy of $\mathbb{Q}$ in $\mathbb{R}$ that respects the operations and the order (Theorem 5.4.1); $\mathbb{R}$ is Archimedean (Theorem 5.4.2), $\mathbb{Q}$ is dense in it (Theorem 5.4.3), and the class of the Newton sequence is a positive real number whose square is 2 (Example (The Class of the Newton Sequence)). Bisection with rational ends shows that every non-empty set of real numbers that is bounded above has a least upper bound, so $\mathbb{R}$ is a complete ordered field (Theorem 5.5.2), and the proof uses no axiom of choice (Example (Picking Representatives Is a Use of Choice)).

What is open: is this the only complete ordered field? Other constructions give other sets, the cuts of *Axiomatic Set Theory* among them, and Chapter 6 proves that any two complete ordered fields are linked by exactly one order-preserving field isomorphism. The sequence form of completeness, that every Cauchy sequence of real numbers converges, is proved in *Analysis on the Real Line*, Chapter 4.

5.-1.5 Terms

- **Cauchy sequences of rationals**: the set $\mathcal{C}$ of sequences $\mathbb{N} \to \mathbb{Q}$ that are Cauchy in $\mathbb{Q}$ (Definition 5.1.1).
- **Cauchy-complete**: said of an ordered field in which every Cauchy sequence converges (Definition 5.5.1).
- **complete ordered field**: an ordered field in which every non-empty subset that is bounded above has a supremum (Definition 5.5.1).
- **copy of $\mathbb{Q}$ in $\mathbb{R}$**: $\kappa(q) = [(q)_n]$, the class of the constant sequence (Definition 5.4.1).
- **equivalent** (Cauchy sequences): $a \sim b$ when $a - b$ is null (Definition 5.1.2).
- **integer part** (in $\mathbb{R}$): the one $m \in \mathbb{Z}$ with $\kappa(m) \leq x < \kappa(m+1)$ (Proposition 4.2.4 with $F = \mathbb{R}$).
- **positive** (Cauchy sequence): from some index on, every term is at least one fixed positive rational (Definition 5.3.1).
- **real number**: a class $[a]$ of Cauchy sequences of rationals under $\sim$; $\mathbb{R} = \mathcal{C}/\sim$ (Definition 5.2.1).
- **representative**: any $b \in [a]$ (Definition 5.2.1).
- **supremum**: the least upper bound of a set, as defined in *Sets and Functions* (Sets and Functions, Definition 5.3.5).